



Day Trade Scanner Pro

User Guide & Reference Manual

Version 1.0 · June 2026

Platform: Windows 10 / 11 · **Runtime:** .NET 10 (self-contained) · **Architecture:** x64

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Overview

What It Does

Day Trade Scanner Pro monitors the US equity market in real time, scores every symbol by a composite **Trade Score**, and surfaces the highest-probability trade candidates on your screen before the move is over. It replaces the manual process of scanning dozens of charts every morning.

The app connects to up to seven market-data providers simultaneously. If one fails or rate-limits, it automatically fails over to the next. Premium providers (Polygon.io, Finnhub) unlock live WebSocket price streaming; free providers (Yahoo Finance, Stooq) work with no configuration at all.

Feature Summary



Automated Scanning

Continuous or one-shot scans with configurable frequency (1 min → 1 hour). Results ranked by Trade Score, highest first.



Trade Score 0–100

Composite score combining Momentum, Volume, Technical indicators, and News sentiment. Weights are fully adjustable.



Live WebSocket Streaming

Real-time trade ticks from Polygon.io and Finnhub. Prices update in the results table without re-scanning. Auto-reconnects on drops.



Alerts & Notifications

Price above/below, percent change, and volume spike alerts with sound and popup notifications.



Candlestick Charts

Interactive OHLCV charts for any scanned symbol using LiveCharts2. Multiple timeframes supported.



Export

One-click export to formatted `.xlsx` (color-coded scores, auto-filter) or plain `.csv`.



Watchlist

Maintain a persistent list of symbols to monitor. Watchlist symbols are included in every scan run.



Encrypted Key Storage

API keys stored with Windows DPAPI — encrypted at rest, bound to your Windows account. Never stored in plaintext.

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Installation

System Requirements

Component	Minimum	Recommended
Operating System	Windows 10 (build 1809)	Windows 11
Architecture	x64	x64
RAM	4 GB	8 GB+
Disk	200 MB free	500 MB (for scan history)
Internet	Required (market data)	Broadband
.NET Runtime	Bundled — no pre-install required	

Installing the MSIX Package

1

Download the installer

Download `TradeScanner_1.0.0.0.msix` from the [GitHub Releases](#) page.

2

Install the signing certificate (sideload only)

If this is a self-signed build, install the certificate first. Open PowerShell as Administrator:

```
Import-Certificate -FilePath TradeScanner-test.pfx -CertStoreLocation Cert:\LocalMachine\TrustedPeople
```

3

Enable Developer Mode or sideloading

Go to **Settings → System → For developers** and enable *Sideload apps* (Windows 10) or *Developer Mode* (Windows 11).

4

Run the installer

Double-click `TradeScanner_1.0.0.0.msix`. Click **Install** in the App Installer dialog.

5

Launch the app

Find **Day Trade Scanner Pro** in the Start menu. On first launch the database is created automatically.

NOTE

The app stores its database at `%LOCALAPPDATA%\TradeScanner\tradescanner.db`. This file is created on first run and persists scan history, watchlists, alerts, and encrypted API keys.

Uninstalling

Go to **Settings** → **Apps** → **Installed apps**, find *Day Trade Scanner Pro*, and click **Uninstall**. User data (database file) at `%LOCALAPPDATA%\TradeScanner\` is not removed automatically — delete that folder to fully clean up.

3 Quick Start

The fastest path to your first scan — no API key required (uses Yahoo Finance and Stooq).

- 1

Launch Day Trade Scanner Pro
The Dashboard opens by default. The status bar shows "Ready".
- 2

Click "Scan Once"
A single scan runs immediately against Yahoo Finance (always available, no key). Results appear in the table within 30–60 seconds depending on symbol list size.
- 3

Review results
Results are sorted by Trade Score (highest first). Each row shows symbol, score, price, change %, volume, and volume ratio.
- 4

Click a symbol to open its chart
Navigate to the **Charts** tab. The selected symbol's candlestick chart loads automatically.
- 5

Add an API key for better data (optional)
Go to **Providers**, paste your Polygon.io or Finnhub key, click **Save**. The provider becomes active immediately — no restart needed.
- 6

Start continuous scanning
Click the green **Start Scan** button. The scanner runs on the interval configured in Settings (default: 5 minutes).

4

Configuring Data Providers

Provider Reference

PROVIDER	TIER	KEY REQUIRED	WEBSOCKET STREAMING	DATA FREQUENCY
Polygon.io	PREMIUM	Yes	YES	Real-time + historical
Finnhub	PREMIUM	Yes	YES	Real-time + historical
Financial Modeling Prep	FREEMIUM	Yes	NO	Quote + historical
TwelveData	FREEMIUM	Yes	NO	Quote + historical
AlphaVantage	FREEMIUM	Yes (free tier available)	NO	Quote + historical
Stooq	FREE	No	NO	Daily bars only
Yahoo Finance	FREE	No	NO	Quote + historical

RECOMMENDATION

For the best experience, add a free Finnhub key (finnhub.io — free tier available) or a Polygon.io Starter key. Both unlock real-time WebSocket streaming that keeps prices updating between scans.

Adding an API Key

- 1

Navigate to Providers

Click **Providers** in the left sidebar.
- 2

Locate your provider

Each provider appears as a card. A green dot means it is currently active; grey means unavailable (no key or key invalid).
- 3

Paste your key

Click the text field labeled *"Paste API key here, then Save or Validate"* and paste your key.
- 4

Click Save

The key is encrypted with Windows DPAPI and stored in the local database. The field clears and a  icon appears — confirming the key is saved.
- 5

Optionally click Validate

Validates the key by making a live test request to the provider. Shows "Valid ✓" or an error message in the status column.

To remove a key, click **Clear**. The provider reverts to unavailable until a new key is entered.

Failover Chain

The app uses an automatic failover chain ordered by priority:

1. Polygon.io (priority 1, fastest)
2. Finnhub (priority 2)
3. Financial Modeling Prep (priority 3)
4. TwelveData (priority 4)
5. AlphaVantage (priority 5)
6. Stooq (priority 6, daily only)
7. Yahoo Finance (priority 10, always available)

If a provider returns an error or rate-limit response, the chain automatically retries with the next available provider. You do not need to configure this — it is automatic.

5

Dashboard

The Dashboard is the main workspace. It shows the scan results table and the scanning/streaming controls.

Running a Scan

BUTTON	ACTION
Start Scan (green)	Starts continuous scanning on the configured interval. Button label changes to <i>Stop Scan</i> . A spinner shows in the status bar while a scan is running.
Scan Once	Runs a single scan immediately and shows results. Does not start continuous scanning.
Export CSV	Opens a save dialog to export current results as <code>.xlsx</code> or <code>.csv</code> .

The status bar (bottom of the toolbar) shows the current state: *Ready*, *Scanning...*, *Scan complete — N results*, or any error message from the provider chain.

Results Table Columns

COLUMN	DESCRIPTION
Symbol	Ticker symbol (e.g., AAPL)
Score	Trade Score 0–100. Higher is a stronger setup. Color-coded: teal ≥ 70 , amber 40–69, red < 40 .
Price	Last trade price. Updates live when streaming is active.
Change %	Intraday change from previous close. Green = positive, red = negative.
Volume	Volume traded today.
Vol Ratio	Today's volume ÷ average daily volume. Values above 2× indicate unusual activity.
Provider	Which data provider supplied this result.
Time	Timestamp of when this symbol was last scanned.

Understanding the Trade Score

The Trade Score is a composite 0–100 rating. Four sub-scores are weighted and summed:

Momentum		35%
Volume		30%
Technical		25%
News		10%

Sub-Score	What Drives It
Momentum	Intraday % change, direction, RSI position (sweet spot: 50–70)
Volume	Volume ratio vs. historical average (5× avg = 100 pts)
Technical	Breakout detection, Bollinger Band position, SMA trend, ATR range
News	Average sentiment of recent news headlines (neutral = 50)

TIP

You can adjust the weights in Settings → Score Weights to match your personal trading style. For pure momentum traders, increase Momentum to 50% and reduce Technical. For swing traders, increase Technical and reduce Momentum.

Real-Time Streaming

Streaming connects a WebSocket to Polygon.io or Finnhub and pushes live trade ticks directly into the results table — prices update without re-scanning.

- 1

Select a streaming provider

Use the dropdown in the toolbar (Polygon or Finnhub). Requires a saved API key for the selected provider.
- 2

Optional: enter API key inline

If you haven't saved a key via the Providers page, paste it in the streaming key field. If the field is left blank, the app silently uses the saved key.
- 3

Click Stream

The button label changes to *Stop Stream*. A "Connected" status appears. Prices in the table begin updating in real time.
- 4

Click Stop Stream to disconnect

The connection closes gracefully. Prices revert to scan-time values.

RECONNECTION

If the WebSocket drops (network interruption, server restart), the app automatically reconnects with exponential backoff: 2s → 4s → 8s → 16s → 32s → 60s. The status bar shows "Connection lost — reconnecting in Xs (attempt N)...". No action is required.

Exporting Results

Click **Export CSV** in the Dashboard toolbar. A Save dialog opens with two file type options:

- Excel Workbook (.xlsx)** — Formatted spreadsheet with bold headers, color-coded Trade Score column (teal ≥ 70, amber 40–69, red < 40), color-coded Change % (teal = positive, red = negative), auto-filter on all columns, and auto-sized column widths.
- CSV File (.csv)** — Plain comma-separated values, compatible with any spreadsheet application.

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Alerts

Alerts notify you when a specific condition is met for a symbol during a scan. All alerts are evaluated at the end of each scan run.

Alert Types

Alert Type	Triggers When	Threshold Meaning
Price Above	Last price > threshold	Dollar value (e.g., 150.00)
Price Below	Last price < threshold	Dollar value (e.g., 100.00)
Percent Change Up	Intraday change % > threshold	Percent value (e.g., 5 = +5%)
Percent Change Down	Intraday change % < -threshold	Percent value (e.g., 3 = -3%)
Volume Above	Current volume > threshold	Share count (e.g., 1000000)

Creating an Alert

- 1

Navigate to Alerts

Click Alerts in the left sidebar.
- 2

Fill in the add-alert form

Enter the ticker symbol (e.g., AAPL), choose an alert type from the dropdown, and enter the threshold value.
- 3

Click Add Alert

The alert appears in the list below, enabled by default.
- 4

Manage existing alerts

Use Toggle to temporarily enable/disable an alert without deleting it. Use Delete to remove it permanently.

The alert list columns show: Symbol, Type, Threshold, Enabled status, how many times it has triggered (*Triggered*), and when it last fired (*Last At*).

Notifications

When an alert fires, the app:

- Plays a system sound (*Asterisk*) if **Sound Alerts** is enabled in Settings

- Shows a popup notification if **Popup Alerts** is enabled in Settings

Both notification types can be toggled independently in **Settings** → **Notifications**.

7 Charts

The Charts page displays a candlestick chart for a selected symbol.

1 Navigate to Charts

Click

Charts

in the left sidebar.

2 Enter a symbol

Type a ticker symbol in the input field (e.g.,) and select a timeframe from the dropdown (5m, 1h, 1d, 1w).

3 Click Load

The chart fetches historical bars from the active provider chain and renders a candlestick series. Green candles = close above open; red candles = close below open.

TIMEFRAME AVAILABILITY

Intraday timeframes (5m, 1h) require a premium provider (Polygon.io, Finnhub, FMP, or TwelveData). Stooq and Yahoo Finance only support daily (1d) and above. If no intraday data is available, the chart shows an empty state.

If no data has been loaded yet, the chart area displays *"No chart data. Enter a symbol and click Load."*. If a load is in progress, a loading spinner is shown instead.

8 Watchlist

The Watchlist lets you maintain a persistent list of symbols you always want scanned, regardless of the general filter criteria.

1 Navigate to Watchlist

Click

Watchlist

in the left sidebar.

2 Add a symbol

Type a ticker in the input field and click

Add

. Duplicates are ignored silently.

3 Remove a symbol

Select a symbol in the list and click

Remove

, or use the delete button on the row.

Watchlist symbols are included in every scan run. They bypass the minimum price, volume, and change-percent filters defined in Settings, so they always appear in results.

PERSISTENCE

Watchlist entries are stored in the local SQLite database and survive app restarts. They are device-specific (not synced across machines).

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Settings

Scanner Settings

SETTING	DESCRIPTION	DEFAULT
Scan Frequency	How often continuous scanning runs: Every1Minute, Every5Minutes, Every15Minutes, Every30Minutes, Hourly	Every5Minutes

Filter Criteria

Filters control which symbols are included in scan results. Watchlist symbols bypass these filters.

FILTER	DESCRIPTION	DEFAULT
Min Price (\$)	Exclude symbols below this price. Use to filter out penny stocks.	\$1.00
Max Price (\$)	Exclude symbols above this price. Use to stay in a tradeable range.	\$500.00
Min Volume	Exclude symbols with volume below this count. Use to ensure liquidity.	100,000
Min Change %	Exclude symbols with intraday change below this percent (absolute value).	1.0%
Min Trade Score	Slider 0–100. Only symbols scoring at or above this value appear in results.	50
Max Results	Maximum number of results to show per scan. Highest scores are kept.	50

Score Weights

Adjust how much each component contributes to the Trade Score. Weights are sliders that display the current percentage. Changes take effect on the next scan run.

WEIGHT SUM

The four weights (Momentum, Volume, Technical, News) should sum to 1.0 (100%). The app does not enforce this — if they don't sum to 1.0, scores will be proportionally scaled. It is your responsibility to keep them balanced.

Notifications

SETTING	DESCRIPTION
Sound Alerts	Play a system sound when an alert condition is met.
Popup Alerts	Show a popup notification when an alert condition is met.

Appearance

The **Theme** dropdown allows switching between available themes. Currently *Dark* is the primary theme; additional themes may be added in future versions.

Click **Save Settings** to persist. Click **Reset Defaults** to revert all settings to factory values.

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Security & Privacy

API Key Storage

API keys are encrypted using **Windows Data Protection API (DPAPI)** before being stored in the local SQLite database. DPAPI binds encryption to your Windows user account and machine — the encrypted blob cannot be decrypted on another machine or under a different Windows account, even if someone copies the database file.

WHAT THIS MEANS IN PRACTICE

If you reinstall Windows, restore your PC to factory settings, or move the database to another computer, saved API keys will not be recoverable. Re-enter them on the Providers page after re-installation.

Data Storage

All application data is stored locally on your device at:

```
%LOCALAPPDATA%\TradeScanner\
  tradescanner.db      - SQLite database (scan history, watchlists, alerts, encrypted keys)
  logs\               - Application log files (Serilog, rotated daily)
```

No data is transmitted to any server other than the market data providers you have configured. The application makes only outbound HTTP/WebSocket requests to provider APIs.

Network Connections

DESTINATION	PROTOCOL	PURPOSE
api.polygon.io	HTTPS / WSS	Market data (if configured)
finnhub.io	HTTPS / WSS	Market data (if configured)
financialmodelingprep.com	HTTPS	Market data (if configured)
api.twelvedata.com	HTTPS	Market data (if configured)
www.alphavantage.co	HTTPS	Market data (if configured)
stoq.com	HTTPS	Free daily data
query1.finance.yahoo.com	HTTPS	Free quote data

11 Troubleshooting

The app won't install (MSIX)

MOST COMMON CAUSE

The signing certificate is not trusted. Run the certificate import command (see Installation, Step 2) from an elevated (Administrator) PowerShell session, then try installing again.

Scan returns no results

- Check that the filter criteria in Settings are not too restrictive. Try lowering Min Trade Score to 0 and Min Change % to 0.
- Verify that at least one provider is available (green dot on the Providers page). If all providers show grey, either no API keys are saved, or Yahoo Finance is failing — check your internet connection.
- Check the status bar for any error message from the provider.

Streaming doesn't connect

- Verify your Polygon.io or Finnhub key is saved on the Providers page ( icon visible).
- Premium WebSocket access may require a paid plan on the provider side. Finnhub's free tier supports WebSocket; Polygon.io requires at least a Starter plan.
- Check for a firewall or VPN blocking outbound WSS connections to `socket.polygon.io` or `ws.finnhub.io`.

Charts show "No chart data"

- Ensure you clicked **Load** after entering the symbol and timeframe.
- If using a 5m or 1h timeframe, ensure a premium provider is configured — Stooq and Yahoo do not support intraday data.
- Outside of US market hours (9:30 AM – 4:00 PM ET), intraday data for today may not yet be available.

API key shows "Invalid" after saving

- Double-check that you copied the entire key with no leading/trailing spaces.
- Some provider free tiers have daily request limits — the validate call may be failing due to rate limiting, not an invalid key.
- If the key was saved from a different Windows account or PC, it will not decrypt correctly. Enter the key again.

Application logs

Detailed logs are written to `%LOCALAPPDATA%\TradeScanner\logs\`, one file per day. These are plain text files useful for diagnosing errors. If reporting a bug, please include the relevant log file.

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Technical Reference

Technical Indicators Computed

INDICATOR	USAGE
RSI (14-period)	Momentum score — sweet spot 50–70 = full 40 pts; overbought ≥ 70 = 20 pts
SMA (20 & 50 period)	Trend confirmation in Technical score
Bollinger Bands (20,2)	Band position for Technical score
ATR (14-period)	Intraday range quality check in Technical score
Volume Ratio	Today's volume ÷ 20-day avg volume; drives Volume score
Breakout Detection	Close above highest high of prior N bars (configurable window)

Database Schema (SQLite)

TABLE	CONTENTS
ScanResults	Historical scan records — all scored symbols per run
WatchlistEntries	User-defined watchlist symbols
AlertRules	User-defined alert conditions and trigger counts
ProviderConfigs	Encrypted API keys per provider
AppSettings	User preferences (scan frequency, filters, weights)
StockSymbols	Master symbol list

Technology Stack

COMPONENT	TECHNOLOGY	VERSION
Platform	.NET	10.0
UI Framework	WPF (Windows Presentation Foundation)	.NET 10
MVVM	CommunityToolkit.Mvvm	8.x
Database	SQLite via EF Core	EF 9.0

Charts	LiveCharts2	rc2
Resilience	Polly	8.x
Excel Export	ClosedXML	0.102.3
Logging	Serilog	4.x
DI Container	Microsoft.Extensions.DependencyInjection	10.0
Installer	MSIX	—